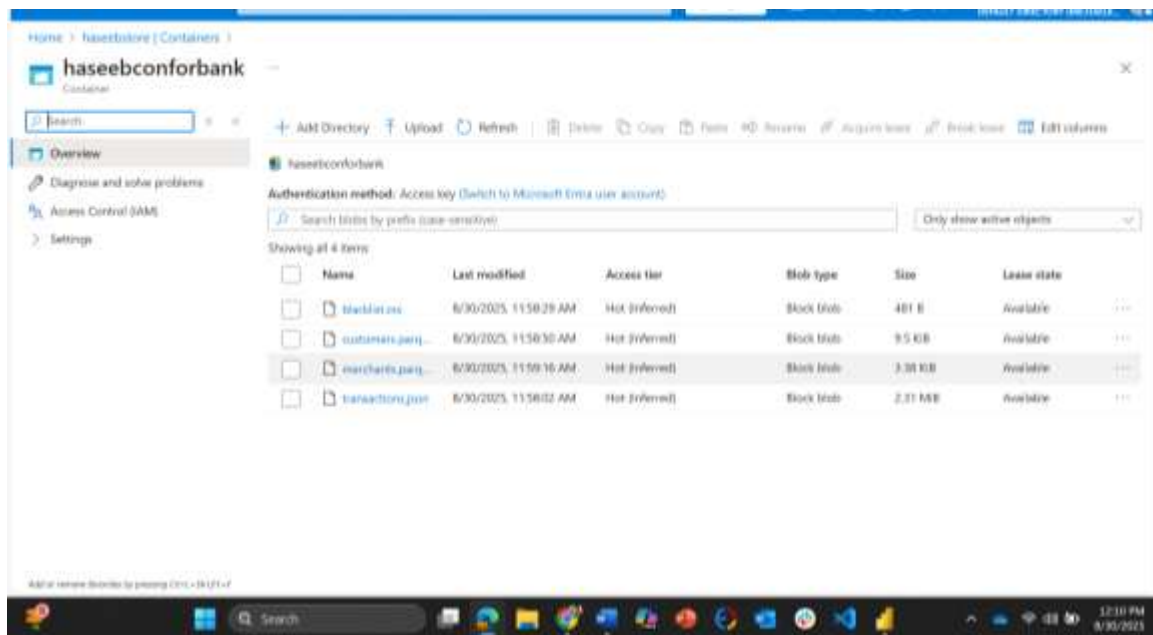
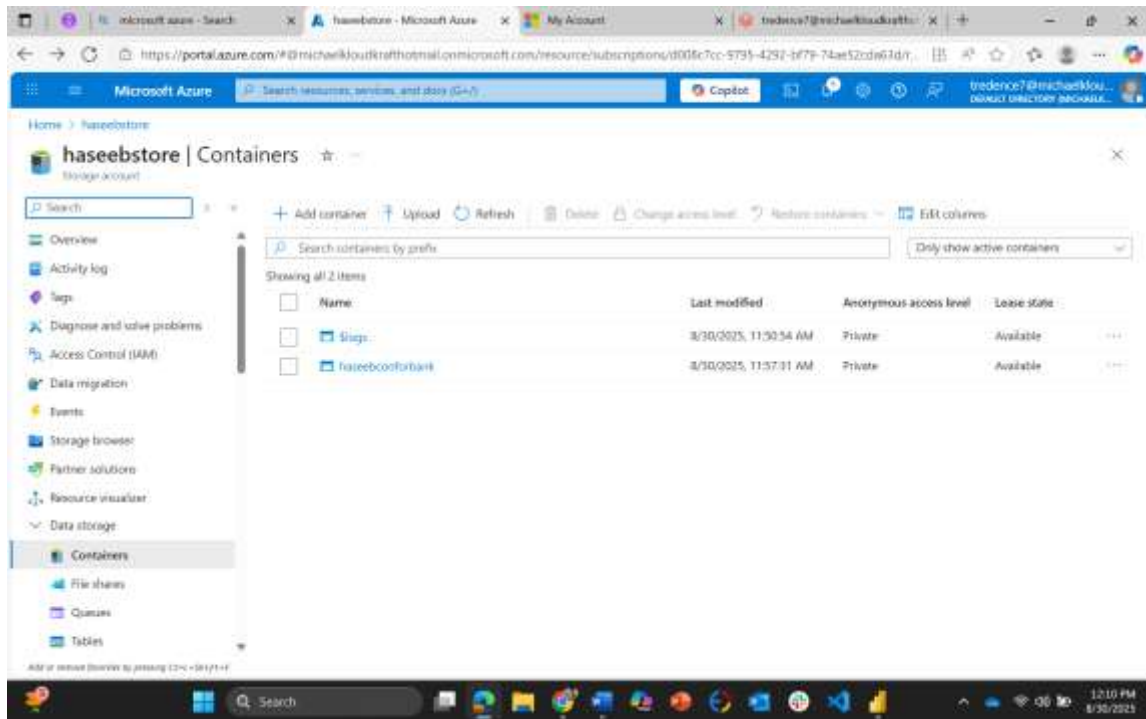


# Haseeb\_Project\_Fraud\_Detection\_in\_Banks

Storage Account:-



## Access Connector:-

The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the 'Microsoft Azure' logo, a search bar, and user account information. The main content area displays the 'haseeb\_access\_connector' resource overview. The left sidebar contains a navigation menu with options like 'Overview', 'Activity log', 'Access control (IAM)', 'Tags', 'Resource visualizer', 'Settings', 'Automation', and 'Help'. The 'Overview' tab is selected, showing details such as 'Resource group (moved)', 'Location', 'Subscription', and 'Subscription ID'. A 'JSON View' button is visible in the top right corner of the overview section.

haseeb\_access\_connector  
Access Connector for Azure Databricks

Search Refresh Delete

Overview

Activity log

Access control (IAM)

Tags

Resource visualizer

Settings

Automation

Help

Essentials

Resource group (moved)  
Labx/ah2022

Location  
Australia Central

Subscription (moved)  
Labx/craft

Subscription ID  
d008c7cc-9795-4292-bf79-74ae52cda63d

Tags (edit)  
Add tags

State  
Succeeded

Resource ID  
/subscriptions/d008c7cc-9795-4292-bf79-74ae52cda63d/resourceGroups/...

JSON View

Copy Copied

App or resource identifier by pressing Ctrl+Shift+F

The screenshot shows the 'Add role assignment' page in the Microsoft Azure portal. The top navigation bar is consistent with the previous screenshot. The main content area is titled 'Add role assignment' and includes tabs for 'Role', 'Members', 'Conditions', and 'Review + assign'. A description of role definitions is provided. The 'Job function roles' section is active, showing a list of roles. A search bar at the top of the list contains the text 'blob'. The roles are displayed in a table with columns for Name, Description, Type, Category, and Details. The 'Storage Blob Data Contributor' role is highlighted. At the bottom of the page, there are buttons for 'Review + assign', 'Previous', and 'Next', along with a 'Feedback' link.

Home > haseebstore | Access Control (IAM)

### Add role assignment

Role Members Conditions Review + assign

A role definition is a collection of permissions. You can use the built-in roles or you can create your own custom roles. [Learn more](#)

Job function roles Privileged administrator roles

Grant access to Azure resources based on job function, such as the ability to create virtual machines.

Search blob Type: All Category: All

| Name ↑                             | Description ↑                                                                                  | Type ↑      | Category ↑ | Details              |
|------------------------------------|------------------------------------------------------------------------------------------------|-------------|------------|----------------------|
| CreateBlobStorage                  |                                                                                                | CustomRole  | None       | <a href="#">View</a> |
| Defender CSPM Storage Data Scan... | Grants access to read blobs and files. This role is used by the data scanner of Defender CSPM. | BuiltInRole | None       | <a href="#">View</a> |
| Defender for Storage Data Scanner  | Grants access to read blobs and update index tags. This role is used by the data scanner of... | BuiltInRole | None       | <a href="#">View</a> |
| Storage Blob Data Contributor      | Allows for read, write and delete access to Azure Storage blob containers and data.            | BuiltInRole | Storage    | <a href="#">View</a> |
| Storage Blob Data Owner            | Allows for full access to Azure Storage blob containers and data, including assigning POS...   | BuiltInRole | Storage    | <a href="#">View</a> |
| Storage Blob Data Reader           | Allows for read access to Azure Storage blob containers and data.                              | BuiltInRole | Storage    | <a href="#">View</a> |
| Storage Blob Delegator             | Allows for generation of a user delegation key which can be used to sign SAS tokens.           | BuiltInRole | Storage    | <a href="#">View</a> |

Review + assign Previous Next

Feedback

## Role Assignment:-

The screenshot displays the Microsoft Azure portal interface for adding a role assignment. The main page is titled "Add role assignment" and includes tabs for "Role", "Members", "Conditions", and "Review + assign". The "Members" tab is active, showing a list of roles with a filter applied: "Showing a filtered list of roles because your permissions include a condition. [Learn more](#) [View all roles](#)". The "Selected role" is "Storage Blob Data Contributor". Under "Assign access to", the "Managed identity" option is selected. The "Members" section shows "No members selected" with a "+ Select members" link. The "Description" field is optional.

A modal window titled "Select managed identities" is open on the right. It displays a warning: "Some results might be hidden due to your ABAC condition." Below this, there are dropdown menus for "Subscription \*" (set to "LabsKraft") and "Managed identity" (set to "Access Connector for Azure Databricks (8)"). A search bar shows "hA". The "Selected members" section lists one member: "haseeb\_access\_connector" with its full path and a "Remove" link. At the bottom of the modal are "Select" and "Close" buttons, and a "Feedback" link.

The Windows taskbar at the bottom shows the time as 12:20 PM on 8/30/2025.

## Catalog Creation:-

The image displays two screenshots from the Databricks web interface, illustrating the process of creating a new catalog and an external location.

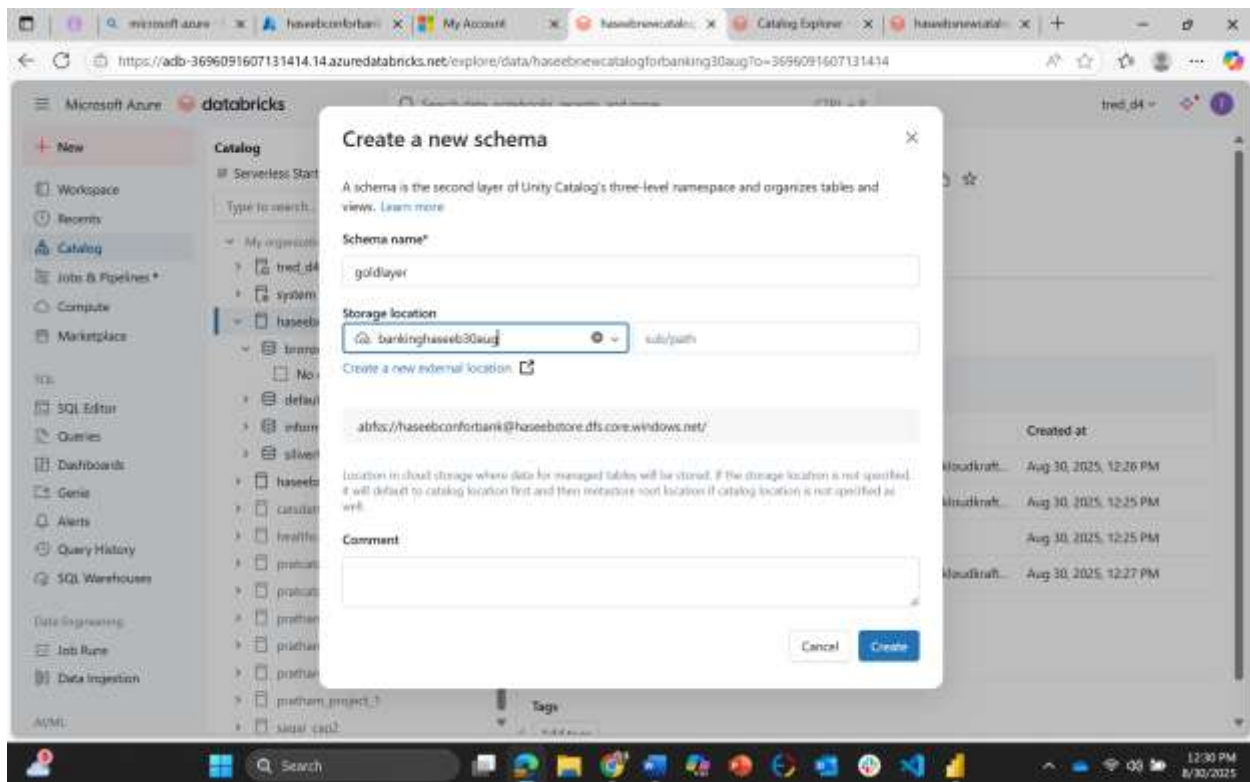
**Top Screenshot: Create a new catalog**

This screenshot shows the 'Create a new catalog' dialog box. The 'Catalog name' field is filled with 'haseebnewcatalogforbanking30aug'. The 'Type' dropdown is set to 'Standard'. The 'Storage location' field is filled with 'abfs://haseebconforbank@haseebstore.dfs.core.windows.net/'. The 'Create a new external location' checkbox is checked. The 'Created at' timestamp is 'Aug 30, 2025, 12:26 PM'.

**Bottom Screenshot: Create external location**

This screenshot shows the 'Create external location' dialog box. The 'External location name' field is filled with 'bankinghaseeb30aug'. The 'Storage type' dropdown is set to 'Azure Data Lake Storage'. The 'URL' field is filled with 'abfs://haseebconforbank@haseebstore.dfs.core.windows.net/'. The 'Storage credential' dropdown is set to 'Create new storage credential'. The 'Access connector ID' field is filled with '/subscriptions/800bc7cc-9795-4292-bf79-74ae52cda63d/resourceGroups/LabKraft2027/providers/Microsoft.Databricks/accessConnectors/haseebconforbank@haseebstore.dfs.core.windows.net/'. The 'User assigned managed identity ID (optional)' field is empty. The 'Comment' field is empty. The 'Created at' timestamp is 'Aug 30, 2025, 12:27 PM'.

## Defining Schema:-





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haseeb\_30aug

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```
spark.conf.set(
    "fs.azure.account.key.haseebstore.dfs.core.windows.net",
    "UoF513Ljag8cA-S+1-iXJr873AQURoBwG1F83DCf3BDH7kFAdH6t1y5pAcEXY8U7B24v+ASck29Q8ss"
)
```

```
df_transactions = spark.read.format("json").option("inferSchema", "true").option("header", "true").load(
    "abfs://haseebconforbank@haseebstore.dfs.core.windows.net/transactions.json"
)
df_customers = spark.read.parquet(
    "abfs://haseebconforbank@haseebstore.dfs.core.windows.net/customers.parquet"
)
df_merchants = spark.read.parquet(
    "abfs://haseebconforbank@haseebstore.dfs.core.windows.net/merchants.parquet"
)
df_blacklist = spark.read.option("header", "true").csv(
    "abfs://haseebconforbank@haseebstore.dfs.core.windows.net/blacklist.csv"
)
```

(4) Spark Jobs

df\_blacklist: myspark sql convert dataframe [DataFrame > lazycollect [it: string, myspark: string]]

12:58 PM 8/30/2025

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```
df_transactions.write.format("delta").mode("overwrite").saveAsTable("haseebnewcatalogforbanking30aug.bronzelayer.transactions_raw")
df_customers.write.format("delta").mode("overwrite").saveAsTable("haseebnewcatalogforbanking30aug.bronzelayer.customers_raw")
df_merchants.write.format("delta").mode("overwrite").saveAsTable("haseebnewcatalogforbanking30aug.bronzelayer.merchants_raw")
df_blacklist.write.format("delta").mode("overwrite").saveAsTable("haseebnewcatalogforbanking30aug.bronzelayer.blacklist_raw")
```

(4) Spark Jobs

Code Text Assistant

```
1 display(spark.sql("SELECT * FROM haseebnewcatalogforbanking30aug.bronzelayer.transactions_raw LIMIT 10"))
2 display(spark.sql("SELECT * FROM haseebnewcatalogforbanking30aug.bronzelayer.customers_raw LIMIT 10"))
3 display(spark.sql("SELECT * FROM haseebnewcatalogforbanking30aug.bronzelayer.merchants_raw LIMIT 10"))
4 display(spark.sql("SELECT * FROM haseebnewcatalogforbanking30aug.bronzelayer.blacklist_raw LIMIT 10"))
```

(6) Spark Jobs

Table +

account id 1.2 amount channel currency device info 1.2 is fraud location

12:58 PM 8/30/2025

Microsoft Azure

dotabricks

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CTRL + P

fred\_d4

New

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

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SQL

SQL Editor

Queries

Dashboards

Genie

Alerts

Query History

SQL Warehouses

Date Engineering

Job Runs

Data Ingestion

AI/ML

Catalog

Serverless Starter Warehouse: Serverless

Type to search...

My organization

fred\_d4\_3696091607131414

system

haseebnewcatalogforbanking30aug

bronze layer

blacklist\_raw

customer\_raw

merchants\_raw

transactions\_raw

default

gold layer

information\_schema

silver layer

haseebnewcatalogforbank

default

haseebbronzaschema

haseebgoldschema

haseebsilverschema

information schema

Catalog Explorer

haseebnewcatalogforbanking30aug

bronze layer

blacklist\_raw

Open in a dashboard

Share

Create

Overview

Sample Data

Details

Permissions

History

Lineage

Insights

Quality

AI Suggested Description

The table contains information related to account-related issues or actions, specifically focusing on the reasons behind certain account activities. It can be used to analyze patterns in account management, understand common issues faced by users, and improve customer support strategies.

Accept

Edit

Send feedback

Filter columns...

AI generate

| Columns    | Type   | Comment | Tags | Column masking ... |
|------------|--------|---------|------|--------------------|
| account_id | string |         |      |                    |
| reason     | string |         |      |                    |

About this table

Owner

Tredence LabsKraft7

Type

Managed

Data source

Delta

12:58 PM

8/30/2025

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```
from pyspark.sql.functions import col, when

# Set Unity Catalog and schema
spark.sql("USE CATALOG hasseebnewcatalogforbanking3Raig")
spark.sql("USE SCHEMA bronze1aig")

DataFrame[]
```

df\_customers = spark.read.table("customers\_raw")

df\_customers: pyspark.sql.connect.dataframe.DataFrame = (account\_id: string, name: string, ... 2 more fields)

```
from pyspark.sql.functions import col, trim, when

df_customers_cleaned = (
    # df_customers
    # 1. Remove NULL account_id
    .filter(col("account_id").isNotNull())
    # 2. Remove INVALID account_ids
    .filter(~col("account_id").startswith("INVALID_"))

```

1:42 PM 8/30/2025

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```
from pyspark.sql.functions import col, trim, when

df_customers_cleaned = (
    # df_customers
    # 1. Remove NULL account_id
    .filter(col("account_id").isNotNull())
    # 2. Remove INVALID account_ids
    .filter(~col("account_id").startswith("INVALID_"))

    # 3. Trim spaces from all string cols
    .withColumn("account_id", trim(col("account_id")))
    .withColumn("name", trim(col("name")))
    .withColumn("demographic", trim(col("demographic")))
    .withColumn("kyc_status", trim(col("kyc_status")))

    # 4. Handle missing values: inputs with 'UNKNOWN'
    .withColumn("name", when(col("name").isNull() | (col("name") == ""), "UNKNOWN").otherwise(col("name")))
    .withColumn("demographic", when(col("demographic").isNull() | (col("demographic") == ""), "UNKNOWN").otherwise(
        col("demographic")))
    .withColumn("kyc_status", when(col("kyc_status").isNull() | (col("kyc_status") == ""), "UNKNOWN").otherwise(col(
        "kyc_status")))

    # 5. Drop duplicates
    .dropDuplicates()

```

1:43 PM 8/30/2025



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31/04 PM 11:06

```
df_customers_cleaned.write.mode("overwrite").format("delta").saveAsTable("hasseebnewcatalogforbanking30aug.silverlayer.customers_raw_processed")
```

2 Spark Jobs

31/05 PM 11:06

```
%%sql
select * from hasseebnewcatalogforbanking30aug.silverlayer.customers_raw_processed
```

2 Spark Jobs

```
%%sql pyspark.sql.connect.dataframe.DataFrame = [account_id string, name string, 2 more fields]
```

| Table       | +          |
|-------------|------------|
| account_id  | name       |
| demographic | kyc_status |
| AC000311    | Cust_SHNYX |
| AC000106    | Cust_BXUO  |
| AC000177    | Cust_H00D4 |
| AC000265    | Cust_B594M |
| AC000297    | Cust_L0D96 |
| AC000299    | Cust_A0YCE |
| AC000336    | Cust_BV1O4 |

3:43 PM 8/30/2025

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31/05 PM 11:06

```
from pyspark.sql.functions import col, when, trim, lower, lit, count

# Read from bronze
df_merchants = spark.read.table("hasseebnewcatalogforbanking30aug.bronzelayer.merchants_raw")

# 1. Remove invalid merchant_id rows
df_merchants = df_merchants.filter(
    (col("merchant_id").isNotNull()) & (~col("merchant_id").startswith("INVALID_"))
)

# 2. Clean mcc
# Find mode of mcc
mode_mcc = df_merchants.filter(
    (~col("mcc").isNotNull()) & (~col("mcc").startswith("INVALID_")) & (trim(col("mcc")) != "")
).groupBy("mcc").count().orderBy(col("count").desc()).first()[0]

df_merchants = df_merchants.withColumn(
    "mcc_imputed",
    when((col("mcc").isNotNull()) & (trim(col("mcc")) == "")) | (col("mcc").startswith("INVALID_")), lit(mode_mcc))
).withColumn(
    "mcc",
    when((col("mcc").isNotNull()) & (trim(col("mcc")) == "")) | (col("mcc").startswith("INVALID_")), lit(mode_mcc))
    ,otherwise(col("mcc"))
)

```

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31-10 PM 12:50 10 Python

```
# 3. Clean risk_rating
# Find mode of risk_rating
mode_risk = df_merchants.filter(
    (~col("risk_rating").isNull()) & (~col("risk_rating").startswith("INVALID_")) & (trim(col("risk_rating")) != "")
).groupBy("risk_rating").count().orderBy(col("count").desc()).first()[0]

df_merchants = df_merchants.withColumn(
    "risk_imputed",
    when((col("risk_rating").isNull()) | (trim(col("risk_rating")) == "")) & (col("risk_rating").startswith("INVALID_")), 1, otherwise(0)
).withColumn(
    "risk_rating",
    when((col("risk_rating").isNull()) | (trim(col("risk_rating")) == "")) & (col("risk_rating").startswith("INVALID_")), lit(mode_risk)
    , otherwise(col("risk_rating"))
)

# 4. Trim and standardize
df_merchants = df_merchants.withColumn("mcc", trim(col("mcc"))).
    .withColumn("risk_rating", lower(trim(col("risk_rating"))))

# Drop duplicates
df_merchants = df_merchants.dropDuplicates()
```

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31-10 PM 1:00 11 Python

```
# Save cleaned merchants into silver
df_merchants.write.mode("overwrite").format("delta").saveAsTable("haseebnewcatalogforbanking30aug.silverlayer.
merchants_raw_processed")
```

31-10 PM 1:04 12 SQL

```
%sql
select * from haseebnewcatalogforbanking30aug.silverlayer.merchants_raw_processed
```

31-10 PM 1:05 13 SQL

```
%sql df = pyspark.sql.connect.dataframe(DataFrame = [merchant_id: string, mcc: string ... 3 more fields])
```

Table

|   | merchant_id | mcc  | risk_rating | mcc_imputed | risk_imp... |
|---|-------------|------|-------------|-------------|-------------|
| 1 | MC00161     | 5411 | high        | 0           | 0           |
| 2 | MC00178     | 7995 | high        | 0           | 0           |
| 3 | MC00051     | 5521 | high        | 0           | 0           |
| 4 | MC00178     | 7995 | high        | 0           | 1           |

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haseeb\_30aug

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13 Python

```
# Read blacklist from bronze
df_blacklist = spark.read.table("haseebnewcatalogforbanking30aug.bronze.layer.blacklist_raw")

# Cleaning steps
df_blacklist_cleaned = {
  df_blacklist
  # Remove null or invalid account_id
  .filter((col("account_id").isNotNull()) & (~col("account_id").startswith("INVALID")))

  # Trim spaces
  .withColumn("account_id", trim(col("account_id")))

  # Remove duplicates
  .dropDuplicates()
}

# Save to silver schema
df_blacklist_cleaned.write.mode("overwrite").format("delta").saveAsTable("haseebnewcatalogforbanking30aug.silver.layer.blacklist_raw_processed")
```

(2) Spark Jobs

df\_blacklist: pyspark.sql.connect.dataframe.DataFrame = (account\_id: string, reason: string)

df\_blacklist\_cleaned: pyspark.sql.connect.dataframe.DataFrame = (account\_id: string, reason: string)

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haseeb\_30aug

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14 SQL

```
select * from haseebnewcatalogforbanking30aug.silver.layer.blacklist_raw_processed
```

(2) Spark Jobs

\_sqldf\_ pyspark.sql.connect.dataframe.DataFrame = (account\_id: string, reason: string)

Table +

|    | account_id | reason      |
|----|------------|-------------|
| 6  | AC000008   | stolen_card |
| 7  | AC000095   | stolen_card |
| 8  | AC000085   | stolen_card |
| 9  | AC000125   | stolen_card |
| 10 | AC000409   | stolen_card |
| 11 | AC000451   | stolen_card |
| 12 | AC000407   | stolen_card |
| 13 | AC000372   | stolen_card |
| 14 | AC000389   | stolen_card |
| 15 | AC000362   | stolen_card |
| 16 | AC000105   | stolen_card |
| 17 | AC000031   | stolen_card |
| 18 | AC000378   | stolen_card |

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Date Engineering Job Runs Data Ingestion

haseeb\_30aug

File Edit View Run Help Python Tabs: ON 15 Last edit was 2 minutes ago Interrupt kraft100 Schedule Share

```
from pyspark.sql.functions import col, trim, to_timestamp

# Load from Bronze table
df_transactions_bronze = spark.read.table("haseebnewcatalogforbanking30aug.bronze.transactions_raw")

# Clean and standardize
df_transactions_silver = (
    df_transactions_bronze
    .dropDuplicates(["transaction_id"]) # remove duplicate transactions
    .na.drop(subset=["transaction_id", "account_id", "amount"]) # drop rows with nulls in key columns
    .withColumn("timestamp", to_timestamp("timestamp")) # ensure timestamp format
    .withColumn("amount", col("amount").cast("double")) # keep amount as double
    .withColumn("is_fraud", col("is_fraud").cast("int")) # cast fraud flag to int (0/1)
    .withColumn("account_id", trim(col("account_id")))
    .withColumn("merchant_id", trim(col("merchant_id")))
    .withColumn("location", trim(col("location")))
    .withColumn("device_info", trim(col("device_info")))
    .withColumn("currency", trim(col("currency")))
    .filter(col("amount") > 0) # remove invalid negative/zero transactions
)

# Save as Silver Delta Table
df_transactions_silver.write.format("delta").mode("overwrite") \
    .saveAsTable("haseebnewcatalogforbanking30aug.silver.transactions_cleaned")
```

1:44 PM 8/30/2025

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Date Engineering Job Runs Data Ingestion

Catalog Serverless Starter Warehouse: Serverless

Type to search...

- blackint\_raw
- customers\_raw
- merchants\_raw
- transactions\_raw
- default
- goldlayer
- information\_schema
- silverlayer
  - blackint\_raw\_processed
  - customers\_raw\_processed
  - merchants\_raw\_processed
  - transactions\_cleaned
- haseebnewcatalogforbank
- csdata
- healthbank\_silver\_catalog
- protocatalog
- protocatalog1
- protham\_02\_project

|                |           |
|----------------|-----------|
| currency       | string    |
| device_info    | string    |
| is_fraud       | int       |
| location       | string    |
| merchant_id    | string    |
| timestamp      | timestamp |
| transaction_id | string    |

silverlayer haseebnewcatalogforbanking30aug

Popularity

Tags

Add tags

Row filter

Add filter

1:52 PM 8/30/2025

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+ New

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie

Alerts

Query History

SQL Warehouses

Date Engineering

Job Runs

Data Ingestion

AI/ML

File Edit View Run Help Python Tabs: ON Last edit was 2 minutes ago Interrupt kraft100 Schedule Share

df.transactions\_clean: pyspark.sql.connect.dataframe.DataFrame = [account\_id: string, amount: double ... 8 more fields]

|   | account_id | amount  | channel   | currency | device_info | is_fraud | location |
|---|------------|---------|-----------|----------|-------------|----------|----------|
| 1 | AC0000043  | 1289.03 | MobileApp | INR      | Device_8354 | 0        | London   |
| 2 | AC0000294  | 1903.39 | ATM       | USD      | Device_46AT | 0        | Tokyo    |
| 3 | AC0000201  | 341.85  | Online    | JPY      | Device_5D3G | 0        | Berlin   |
| 4 | AC0000095  | 801.1   | MobileApp | GBP      | Device_7X2E | 0        | Paris    |
| 5 | AC0000001  | 1223.11 | MobileApp | EUR      | Device_71C8 | 0        | Paris    |
| 6 | AC0000021  | 10000   | POS       | JPY      | Device_9YMS | 1        | Tokyo    |
| 7 | AC0000340  | 191.39  | Online    | GBP      | Device_X0F3 | 0        | LA       |
| 8 | AC0000227  | 110.08  | MobileApp | JPY      | Device_45CR | 0        | LA       |
| 9 | AC0000003  | 662.46  | MobileApp | EUR      | Device_0HQ2 | 0        | Mumbai   |

10 rows | 26.24s runtime Refreshed 4 minutes ago

Waiting 16

Verify

```
display(spark.sql("SELECT * FROM haseebnewcatalogforbanking3aug.silverlayer.transactions_cleaned LIMIT 40"))
```

1:44 PM 8/30/2025



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17 hours ago 30

```
SQL
CREATE OR REPLACE TABLE hassebnwecatalogforbanking18aug.silver_secure.customers_masked AS
SELECT
  account_id, -- keep
  regexp_replace(name, '[.]', 'XXXX') AS name,
  demographic, -- can keep (unless it contains sensitive info like age, gender etc.)
  kyc_status
FROM hassebnwecatalogforbanking18aug.silver_layer.customers_raw_processed;
```

num\_affected\_rows num\_inserted\_rows

No rows returned

0 rows | 9.80s runtime Refreshed 17 hours ago

This result is stored as \_sqlidf and can be used in other Python and SQL cells.

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17 hours ago 31

```
SQL
select * from hassebnwecatalogforbanking18aug.silver_secure.customers_masked limit 10
```

num\_affected\_rows num\_inserted\_rows

|    | account_id | name       | demographic | kyc_status  |
|----|------------|------------|-------------|-------------|
| 1  | AC000047   | XXXXXXXXXX | Adult       | KYC_OK      |
| 2  | AC000270   | XXXXXXXXXX | Youth       | KYC_OK      |
| 3  | AC000024   | XXXXXXXXXX | Adult       | KYC_BLOCKED |
| 4  | AC000295   | XXXXXXXXXX | Adult       | KYC_BLOCKED |
| 5  | AC000345   | XXXXXXXXXX | Youth       | KYC_BLOCKED |
| 6  | AC000254   | XXXXXXXXXX | Adult       | KYC_BLOCKED |
| 7  | AC000303   | XXXXXXXXXX | Senior      | KYC_PENDING |
| 8  | AC000522   | XXXXXXXXXX | Youth       | KYC_PENDING |
| 9  | AC000400   | XXXXXXXXXX | Senior      | KYC_BLOCKED |
| 10 | AC000261   | XXXXXXXXXX | Adult       | KYC_OK      |

10 rows | 4.89s runtime Refreshed 17 hours ago

This result is stored as \_sqlidf and can be used in other Python and SQL cells.

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Hasseeb01

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10 rows | 4.88s runtime Refreshed 17 hours ago

This result is stored as `_sqldf` and can be used in other Python and SQL cells.

```
SQL
CREATE OR REPLACE TABLE hasseebnewcatalogforbanking36aug.silver_secure.transactions_masked AS
SELECT
  transaction_id,
  account_id,
  merchant_id,
  timestamp,
  0 AS amount, -- mask numeric with 0
  currency,
  regexp_replace(location, '(.*)', 'XXX') AS location,
  channel,
  regexp_replace(device_info, '(.*)', 'XXX') AS device_info,
  is_fraud
FROM hasseebnewcatalogforbanking36aug.silver_lwyer.transactions_raw_processed limit 10
```

num.affected.rows num.inserted.rows

10:28 AM 9/1/2025

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```
# Create gold layer tables for analytics
from pyspark.sql.functions import col, sum, count, when, avg, date_trunc

# Daily fraud summary by channel
df_daily_fraud = spark.table("haseebnewcatalogforbanking18aug.silverlayer.silver_final")

daily_fraud_summary = (df_daily_fraud
    .withColumn("date", date_trunc("day", col("timestamp_clean")))
    .groupBy("date", "channel", "risk_rating")
    .agg(
        sum("amount").alias("total_amount"),
        sum("is_fraud").alias("fraud_count"),
        count("*").alias("total_transactions"),
        (sum("is_fraud") / count("*") * 100).alias("fraud_percentage")
    )
    .orderBy("date", "channel"))

daily_fraud_summary.write.format("delta").mode("overwrite")
    .saveAsTable("haseebnewcatalogforbanking18aug.goldlayer.daily_fraud_summary")

# Merchant risk analysis
merchant_risk_analysis = (df_daily_fraud
    .groupBy("merchant_id", "risk_rating", "acc")
    .agg(
        sum("amount").alias("total_processed_amount"),
        sum(when(col("is_fraud") == 1, col("amount"))).alias("fraud_amount"),
        count("*").alias("total_transactions"),
        sum("is_fraud").alias("fraud_count"),
        (sum("is_fraud") / count("*") * 100).alias("fraud_rate")
    )
    .orderBy("fraud_rate", ascending=False))

merchant_risk_analysis.write.format("delta").mode("overwrite")
    .saveAsTable("haseebnewcatalogforbanking18aug.goldlayer.merchant_risk_analysis")
```

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```
daily_fraud_summary.write.format("delta").mode("overwrite")
    .saveAsTable("haseebnewcatalogforbanking18aug.goldlayer.daily_fraud_summary")

# Merchant risk analysis
merchant_risk_analysis = (df_daily_fraud
    .groupBy("merchant_id", "risk_rating", "acc")
    .agg(
        sum("amount").alias("total_processed_amount"),
        sum(when(col("is_fraud") == 1, col("amount"))).alias("fraud_amount"),
        count("*").alias("total_transactions"),
        sum("is_fraud").alias("fraud_count"),
        (sum("is_fraud") / count("*") * 100).alias("fraud_rate")
    )
    .orderBy("fraud_rate", ascending=False))

merchant_risk_analysis.write.format("delta").mode("overwrite")
    .saveAsTable("haseebnewcatalogforbanking18aug.goldlayer.merchant_risk_analysis")
```

Spark Jobs

- daily\_fraud\_summary: pyspark.sql.connect.dataframe.DataFrame = [date: timestamp, channel: string ... 5 more fields]
- df\_daily\_fraud: pyspark.sql.connect.dataframe.DataFrame = [account\_id: string, merchant\_id: string ... 16 more fields]
- merchant\_risk\_analysis: pyspark.sql.connect.dataframe.DataFrame = [merchant\_id: string, risk\_rating: string ... 8 more fields]

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SQL

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SQL

```
select * from haseebnewcatalogforbanking30aug.goldlayer.merchant_risk_analysis limit 20
```

Table

|    | merchant_id | risk_rating | mc   | 1.2 total_processed_amount | 1.2 fraud_amount | 2 total_transactions |
|----|-------------|-------------|------|----------------------------|------------------|----------------------|
| 7  | MC00113     | medium      | 7011 | 163985.05364435713         | 14999.99         |                      |
| 8  | MC00139     | low         | 5812 | 48434.000000000015         | 0.00             |                      |
| 9  | MC00103     | medium      | 5921 | 175046.35364435718         | 0.00             |                      |
| 10 | MC00023     | low         | 6011 | 174161.56364435705         | 10000            |                      |
| 11 | MC00197     | low         | 6011 | 50555.63000000001          | 14566.99         |                      |
| 12 | MC00096     | high        | 1812 | 158517.1536443571          | 9999.99          |                      |
| 13 | MC00206     | low         | 5411 | 278752.2972887142          | 6025.65          |                      |
| 14 | MC00060     | high        | 6011 | 157684.62364435708         | 0.00             |                      |
| 15 | MC00003     | low         | 7995 | 263096.7472887142          | 0.00             |                      |
| 16 | MC00147     | low         | 5921 | 266454.63728871424         | 11025.66         |                      |
| 17 | MC00163     | medium      | 5411 | 155164.81364435708         | 10000            |                      |
| 18 | MC00150     | high        | 6011 | 166026.11364435707         | 14999.98         |                      |
| 19 | MC00155     | medium      | 6011 | 163462.56364435713         | 0.00             |                      |

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SQL

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SQL

merchant\_risk\_analysis

Open in a dashboard Share Create

Overview Sample Data Details Permissions History Lineage Insights Quality

AI Suggested Description

The table contains data related to merchant transactions and associated risk metrics. It includes information such as the total amount processed by each merchant, the amount flagged as fraudulent, and the total number of transactions. This data can be used to assess the risk levels of different merchants, analyze fraud trends, and evaluate the effectiveness of fraud prevention measures.

Accept Edit Send feedback

Filter columns...

AI generate

| Column                 | Type   | Comment | Tags | Column masking ... |
|------------------------|--------|---------|------|--------------------|
| merchant_id            | string |         |      |                    |
| risk_rating            | string |         |      |                    |
| mc                     | string |         |      |                    |
| total processed amount | double |         |      |                    |



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### Settings

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### Developer

Manage your development settings

#### Access tokens

Tokens are disabled for your organization as you do not have permissions to use them. Please contact your administrator for more information.

#### SQL query snippets

Configure SQL query snippets. Note: SQL query snippets will be moving to the SQL editor dropdown menu.

SQL Editor

#### Editor settings

##### General

#### Spark tips

Enrich notebook error stack traces by displaying high-level "error hints" which explains otherwise confusing errors.

On

#### Databricks Advisor

Databricks Advisor displays context aware optimization hints in the notebooks. [Learn more](#)

On

